NORTHERN UNIVERSITY
Knowledge for Innovation and Change

PROJECT PROPOSAL

MEDICONNECT: DOCTOR APPOINTMENT MANAGEMENT SYSTEM

COURSE : DATABASE MANAGEMENT SYSTEM (DBMS) LAB
DEPARTMENT : COMPUTER SCIENCE & ENGINEERING (CSE)

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PROJECT OVERVIEW

The healthcare sector increasingly relies on digital systems to improve efficiency and patient experience. Traditional appointment booking methods often involve manual record keeping, scheduling conflicts, and long waiting times. To address these issues, this project proposes the development of a web-based Doctor Appointment Management System.

The system will enable patients to book appointments with doctors, while administrators can manage doctor information, patient records, and appointment schedules through a centralized database.

PROBLEM STATEMENT

Many small clinics and healthcare centers still manage appointments manually. This leads to:

- Data redundancy and inconsistency
- Difficulty in tracking patient records
- Appointment scheduling conflicts
- Increased administrative workload
- Lack of centralized data management

A computerized database system can effectively solve these problems.

OBJECTIVES

The main objectives of this project are:

- To develop a web-based appointment management system.
- To maintain patient and doctor information in a structured database.
- To facilitate appointment booking and management.
- To reduce manual record-keeping errors.
- To demonstrate the practical implementation of DBMS concepts.

SCOPE OF THE PROJECT

The system will provide the following functionalities:

Patient Management

- Add new patients
- View patient information
- Update patient details
- Delete patient records

Doctor Management

- Add doctors
- Manage doctor profiles
- Store specialization and availability information

Appointment Management

- Book appointments
- View appointment history
- Update appointment status
- Cancel appointments

Reporting

- Display all appointments
- View patient and doctor lists

TECHNOLOGY STACK

Component	Technology
Frontend	HTML, CSS
Backend	PHP
Database	MySQL
Development Environment	XAMPP
Database Tool	phpMyAdmin
Version Control	Git & GitHub

DATABASE DESIGN

Tables

Patients

- patient_id (Primary Key)
- full_name
- email
- phone
- created_at

Doctors

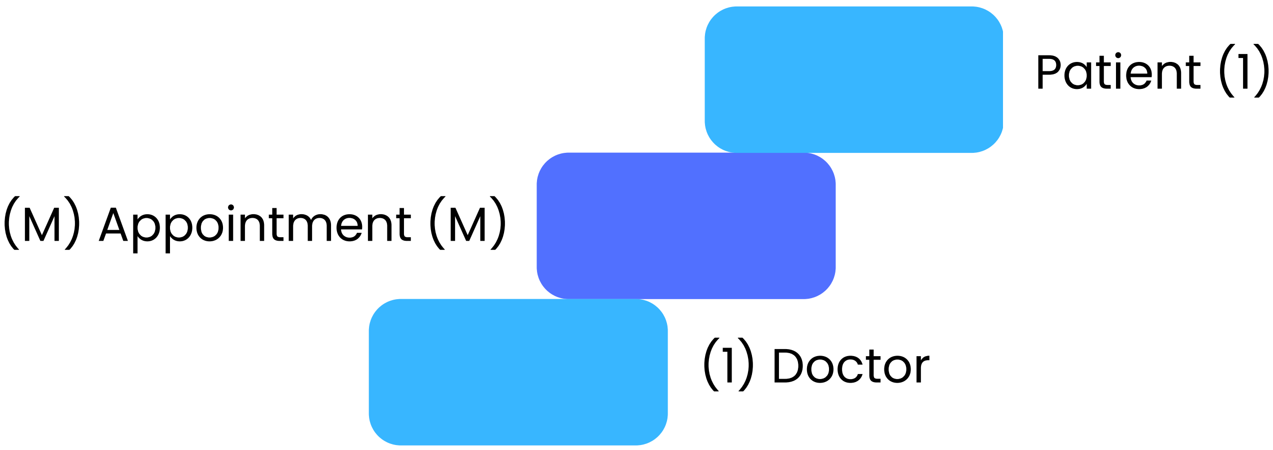
- doctor_id (Primary Key)
- doctor_name
- specialization
- available_time

Appointments

- appointment_id (Primary Key)
- patient_id (Foreign Key)
- doctor_id (Foreign Key)
- appointment_date
- status

ENTITY RELATIONSHIP (ER) OVERVIEW

Relationship Structure:



A patient can book multiple appointments, and a doctor can receive multiple appointments.

EXPECTED OUTCOMES

Upon completion, the system will:

- Store patient information securely.
- Manage doctor schedules efficiently.
- Allow appointment booking and tracking.
- Demonstrate CRUD operations.
- Implement relational database concepts such as primary keys, foreign keys, and SQL joins.

DBMS CONCEPTS USED

- Database Creation
- Tables and Attributes
- Primary Keys
- Foreign Keys
- Relationships
- SQL Queries
- CRUD Operations
- JOIN Operations
- Data Integrity Constraints

FUTURE ENHANCEMENTS

The system can be further improved by adding:

- User Authentication System
- Email Notifications
- Online Payment Gateway
- Prescription Management
- Medical History Tracking
- Doctor Dashboard
- Patient Dashboard
- Mobile Application Integration

CONCLUSION

The Doctor Appointment Management System is designed to simplify the appointment scheduling process while demonstrating fundamental Database Management System concepts. The project will provide practical experience in designing databases, implementing relationships, and developing a complete web-based application using PHP and MySQL.